

JAVA SDE-2 INTERVIEW PREPARATION SERIES

SERIES INDEX - 3 SEGMENTS / 40 BOOKS

Java SDE-2 Interview Preparation Series Index

A Basics-to-Advanced Navigation Guide

BY

Vinay Reddy Kalluri

Choose Java, DSA, or System Design and Backend, then follow a clear prerequisite-aware path within the segment.

JAVA ENGINEERING | DSA | SYSTEM DESIGN AND BACKEND

JAVA 21 | INTERVIEW CORE | SDE-2 FOLLOW-UPS | PRINTABLE EDITION

Series edition - Release 2026-07-25

Open educational interview-preparation edition

Choose Your Learning Segment

This library has three independent starting points: Java Engineering, Data Structures and Algorithms, and System Design and Backend. Choose the segment that matches your immediate goal, then follow its numbered books in order. You can study a second segment in parallel after its prerequisites are comfortable.

Keep the PDFs together in the same directory so relative links can work in viewers that permit local-file navigation. The printed filenames remain the fallback when a viewer blocks those links.

Choose your starting point in 60 seconds

Start rule: select one segment first. Inside it, begin with Book 01 unless you can pass that book's readiness checks without notes.

Your goal	Start here	Continue with
Learn or rebuild Java	JAVA 01 - Java Foundations	Continue through the Java segment in order
Build interview problem-solving skill	DSA 01 - Time and Space Complexity	Continue through the DSA segment in order
Prepare for Java backend and design rounds	SD 01 - Backend and Design Foundations	Continue through databases, Spring, messaging, and distributed systems

The learning loop for every volume

- 1 **Recognize:** use the entry signals and decision maps to identify the pattern.
- 2 **Understand:** restate the contract, invariant, and failure modes in your own words.
- 3 **Implement:** type the Java example and test boundaries before looking back.
- 4 **Explain:** give the complexity, trade-offs, and one production follow-up aloud.
- 5 **Prove readiness:** pass the completion check before opening the next volume.

Part 1 - Java Engineering

Start with the language, developer workflow, build tools, core libraries, JVM, concurrency, performance, and Java interview revision.

JAVA 01 - Java Foundations for Problem Solving

This volume is the canonical Java language foundation and the recommended first book in the series. It starts with source-to-execution and program structure, builds through types, control flow, methods, arrays, strings, objects, core libraries, exceptions, generics, and usage-first collections, then converts that knowledge into interview-quality implementation and debugging skill.

Open [Java-SDE2-DSA-03-Java-Foundations-for-Problem-Solving.pdf](#).

JAVA 02 - Git and GitHub for Java Engineers - Roadmap Edition

Define the complete learning path for the source-control skills expected from a Java SDE-2 engineer before the chapter set is expanded.

Open [Java-SDE2-JAVA-02-Git-and-GitHub.pdf](#).

JAVA 03 - Maven and Gradle for Java Engineers - Roadmap Edition

Keep both dominant Java build tools together so readers learn shared concepts first and tool-specific mechanics second.

Open [Java-SDE2-JAVA-03-Maven-and-Gradle.pdf](#).

JAVA 04 - Advanced Java B: Language, OOP, and Modern Java

Part B covers the language-engineering depth expected when SDE-2 interviews move beyond coding patterns into API design and type safety.

Open [Java-SDE2-ADV-18B-Language-OOP-and-Modern-Java.pdf](#).

JAVA 05 - Advanced Java C: Collections, Streams, and I/O

Part C is the advanced Java-library companion to the DSA volumes, emphasizing API contracts, representation costs, resource ownership, and production boundaries.

Open [Java-SDE2-ADV-18C-Collections-Streams-and-IO.pdf](#).

JAVA 06 - Advanced Java A: JVM and Execution

Part A builds the execution model needed to reason about Java behavior beneath syntax without presenting implementation details as language guarantees.

Open [Java-SDE2-ADV-18A-JVM-and-Execution.pdf](#).

JAVA 07 - Advanced Java D: Concurrency and the Memory Model

Part D develops concurrency correctness from first principles and then connects it to production scheduling, cancellation, backpressure, and testing.

Open [Java-SDE2-ADV-18D-Concurrency-and-Memory-Model.pdf](#).

JAVA 08 - Advanced Java E: Performance, Diagnostics, and GC Incidents

Part E teaches an evidence-first diagnostic method so performance claims and tuning actions remain falsifiable and operationally safe.

Open [Java-SDE2-ADV-18E-Performance-Diagnostics-and-GC.pdf](#).

JAVA 09 - Advanced Java G: Question Bank, Study Plan, and Reference

Part G is the core Java revision system. It consolidates interview questions, selected solutions, study loops, terminology, and primary references without replacing deliberate practice.

Open [Java-SDE2-ADV-18G-Question-Bank-Study-Plan-and-Reference.pdf](#).

Part 2 - Data Structures and Algorithms

Build complexity intuition, mathematical foundations, implementation patterns, data structures, algorithms, and problem-solving confidence.

DSA 01 - Time and Space Complexity for Java Interviews

This volume is the bridge from Java syntax to algorithmic problem solving. It begins with concrete operation counts, introduces growth and space gently, then develops the qualified cost models, constraints, invariants, amortization, and optimization explanations expected in SDE-2 interviews.

Open [Java-SDE2-DSA-02-Time-and-Space-Complexity.pdf](#).

DSA 02 - Number Systems and Math Foundations for DSA Interviews

This volume closes the mathematical prerequisite gap before bit manipulation, arrays, binary search, hashing, and number-based coding problems. It is practical DSA mathematics, not an academic number-theory textbook.

Open [Java-SDE2-DSA-01-Number-Systems-and-Math-Foundations.pdf](#).

DSA 03 - Number Systems Interview Patterns and Rapid Revision

This linked workbook completes Study Step 03 with the 52-implementation catalog, Java number traps, conceptual and code-output questions, debugging tasks, coding exercises, solutions, follow-up chains, cheat sheet, and readiness assessment.

Open [Java-SDE2-DSA-01B-Number-Systems-Interview-Workbook.pdf](#).

DSA 04 - Bit Manipulation in Java

This dedicated volume turns the binary foundation from Study Step 03A into a progressive Java interview toolkit. It begins with visible bit patterns and safe operators, derives each shortcut instead of asking readers to memorize it, then develops XOR families, subset state, SDE-2 problem patterns, Java APIs, and production-quality bit fields.

Open [Java-SDE2-DSA-04-Bit-Manipulation-in-Java.pdf](#).

DSA 05 - Loop Mastery, Patterns, and Index Calculations

This volume is the bridge from basic Java control flow to reliable array and string problem solving. It begins with the execution model of for, while, do-while, and enhanced-for loops; builds a visual language for ranges and invariants; then derives pointer, compaction, window, search, and matrix patterns without asking the reader to memorize templates blindly.

Open [Java-SDE2-DSA-05-Loop-Mastery-and-Index-Calculations.pdf](#).

DSA 06 - Arrays and Array Problem-Solving Patterns

Arrays are the default interview state space. This volume begins with storage, indexes, references, copying, and traversal before building a constraint-driven pattern map for reliable SDE-2 solutions.

Open [Java-SDE2-DSA-06-Arrays-and-Array-Patterns.pdf](#).

DSA 07 - Strings and String Problem-Solving Patterns

This prerequisite-first volume takes a reader from SDE-1 String mechanics through the substring, frequency, window, parsing, and exact-search patterns expected in SDE-2 interviews. Every advanced technique follows a baseline, invariant, dry run, Java implementation, failure boundary, and complete complexity discussion.

Open [Java-SDE2-DSA-07-Strings-and-String-Patterns.pdf](#).

DSA 08 - Hashing: Maps, Sets, Frequency, and Prefix State

Hashing is both a DSA pattern and a Java object-contract problem. This volume connects the two so expected $O(1)$ claims remain defensible.

Open [Java-SDE2-DSA-08-Hashing-Maps-Sets-and-Prefix-State.pdf](#).

DSA 09 - Recursion and Backtracking in Java

This volume separates recursion as a method contract from backtracking as controlled state-space search, with Java stack and mutation behavior made explicit.

Open [Java-SDE2-DSA-09-Recursion-and-Backtracking.pdf](#).

DSA 10 - Linked Lists: Pointer Reasoning and Mutation

Linked-list problems expose whether a candidate can preserve reachability while mutating references. This volume makes those invariants visible.

Open [Java-SDE2-DSA-10-Linked-Lists.pdf](#).

DSA 11 - Stacks, Queues, Deques, and Monotonic Patterns

This volume unifies ordering data structures around the state each one preserves, then applies monotonic structures to linear-time interview patterns.

Open [Java-SDE2-DSA-11-Stacks-Queues-Deques-and-Monotonic-Patterns.pdf](#).

WHY IT WORKS | DSA 12 - Binary Search: Bounds, Answers, and Invariants

Binary search is a proof pattern over monotone state. This volume centers interval semantics, duplicates, insertion points, overflow-safe midpoint calculation, and search on answers.

Open [Java-SDE2-DSA-12-Binary-Search.pdf](#).

DSA 13 - Trees, Binary Search Trees, and Tries

Tree interviews test recursion, state ownership, structural invariants, and the ability to translate ordering into pruning.

Open [Java-SDE2-DSA-13-Trees-BSTs-and-Tries.pdf](#).

DSA 14 - Heaps, Priority Queues, Selection, and Top-K

This volume distinguishes heap order from global sorting and develops priority-based interview patterns without hiding comparator or mutation risks.

Open [Java-SDE2-DSA-14-Heaps-Priority-Queues-and-Top-K.pdf](#).

DSA 15 - Graphs: Traversal, Ordering, Paths, and Union-Find

Graph problems become manageable once the model and edge semantics are explicit. This volume emphasizes algorithm selection, state, overflow, and correctness.

Open [Java-SDE2-DSA-15-Graphs.pdf](#).

WHY IT WORKS | DSA 16 - Greedy Algorithms: Recognition, Proofs, and Scheduling

Greedy solutions are short to code and hard to justify. This volume centers the proof obligation, not the superficial loop.

Open [Java-SDE2-DSA-16-Greedy-Algorithms.pdf](#).

DSA 17 - Dynamic Programming: State, Transitions, and Optimization

This volume makes dynamic programming a specification process: define exactly what a state means, how it depends on smaller states, and what information can be compressed.

Open [Java-SDE2-DSA-17-Dynamic-Programming.pdf](#).

Part 3 - System Design and Backend

Learn backend boundaries, databases, persistence, Spring, Kafka, caching, distributed systems, and production design in prerequisite order.

SD 01 - Advanced Java F: Design, Backend, Testing, and Security

Part F covers the engineering judgment that distinguishes SDE-2 candidates from syntax-only candidates: boundaries, invariants, failure policy, observability, and maintainability.

Open [Java-SDE2-ADV-18F-Design-Backend-Testing-and-Security.pdf](#).

SD 02 - MySQL for Java Backend Interviews - Roadmap Edition

Establish database fundamentals before ORM and Spring Data abstractions.

Open [Java-SDE2-SD-02-MySQL.pdf](#).

SD 03 - Hibernate and JPA for Java Backend Interviews - Roadmap Edition

Teach ORM behavior only after relational and transaction fundamentals are established.

Open [Java-SDE2-SD-03-Hibernate-and-JPA.pdf](#).

SD 04 - Spring Framework for Java Engineers - Roadmap Edition

Make core Spring behavior explicit before readers use Spring Boot conventions and automation.

Open [Java-SDE2-SD-04-Spring-Framework.pdf](#).

SD 05 - Spring Boot for Java Backend Engineers - Roadmap Edition

Build on Spring Framework mechanics to teach Boot conventions, application boundaries, configuration, and operations.

Open [Java-SDE2-SD-05-Spring-Boot.pdf](#).

SD 06 - Spring Data for Java Backend Engineers - Roadmap Edition

Teach Spring Data after SQL, ORM, Spring Framework, and Spring Boot so generated behavior remains explainable.

Open [Java-SDE2-SD-06-Spring-Data.pdf](#).

SD 07 - MongoDB for Java Backend Interviews - Roadmap Edition

Teach when document storage fits and how schema, indexing, consistency, and access patterns drive the design.

Open [Java-SDE2-SD-07-MongoDB.pdf](#).

SD 08 - Redis for Java Backend Interviews - Roadmap Edition

Teach Redis as a distinct system with explicit consistency and failure boundaries, not merely a fast map.

Open [Java-SDE2-SD-08-Redis.pdf](#).

SD 09 - Apache Kafka and Spring Kafka - Roadmap Edition

Keep Apache Kafka semantics and Spring Kafka integration together while teaching the platform model before framework abstractions.

Open [Java-SDE2-SD-09-Apache-Kafka-and-Spring-Kafka.pdf](#).

SD 10 - Spring Ecosystem Extensions - Roadmap Edition

Capture important Spring ecosystem areas without forcing them prematurely into the core learning sequence.

Open [Java-SDE2-SD-10-Spring-Ecosystem-Extensions.pdf](#).

SD 11 - Spring AI for Java Engineers - Roadmap Edition

Place Spring AI after the core backend and Spring foundations so AI integration is taught as production software engineering.

Open [Java-SDE2-SD-11-Spring-AI.pdf](#).

SD 12 - Advanced Java H: Spring Boot and REST APIs

Part H opens the backend specialist track with the Spring and HTTP reasoning expected in Java SDE-2 interviews, while keeping framework behavior tied to explicit contracts and failure modes.

Open [Java-SDE2-ADV-18H-Spring-Boot-and-REST.pdf](#).

SD 13 - Advanced Java I: Persistence, SQL, JPA, and Caching

Part I develops the persistence judgment needed to design correct, measurable Java data paths instead of treating databases, ORMs, and caches as interchangeable storage APIs.

Open [Java-SDE2-ADV-18I-Persistence-SQL-and-Caching.pdf](#).

SD 14 - Advanced Java J: Distributed Systems and System Design

Part J completes the backend specialist track with a constraint-first system-design method spanning distributed data, messaging, failure control, and operations.

Open [Java-SDE2-ADV-18J-Distributed-Systems-and-System-Design.pdf](#).

Contents

This contents page covers only the current focused PDF. Section-level navigation is available through PDF bookmarks.

CONTENTS NAVIGATION

Choose Your Learning Segment	2
Part 1 - Java Engineering	3
Part 2 - Data Structures and Algorithms	5
Part 3 - System Design and Backend	9
Part IV - Recommended Study Rhythm	13
Build, Practice, Explain, Revisit	13

Part IV - Recommended Study Rhythm

Build, Practice, Explain, Revisit

For each volume, complete one learning pass, one no-notes implementation pass, one spoken explanation pass, and one delayed revision pass. Keep an error log organized by contract, invariant, numeric safety, data-structure choice, complexity, and test coverage. A roadmap edition establishes ordering and scope; return as its chapter set expands rather than treating it as finished instruction.

Series Roadmap

Choose Java, DSA, or System Design and Backend first, then follow the books inside that segment in order. Stable study-step codes remain on filenames and links; the clearer segment book codes appear on the website and every PDF cover.

SEGMENT	BOOK	FOCUSED LEARNING PATH
Java	JAVA 01	Java Foundations for Problem Solving
Java	JAVA 02	Git and GitHub for Java Engineers
Java	JAVA 03	Maven and Gradle for Java Engineers
Java	JAVA 04	Advanced Java B: Language, OOP, and Modern Java
Java	JAVA 05	Advanced Java C: Collections, Streams, and I/O
Java	JAVA 06	Advanced Java A: JVM and Execution
Java	JAVA 07	Advanced Java D: Concurrency and the Memory Model
Java	JAVA 08	Advanced Java E: Performance, Diagnostics, and GC Incidents
Java	JAVA 09	Advanced Java G: Question Bank, Study Plan, and Reference
DSA	DSA 01	Time and Space Complexity for Java Interviews
DSA	DSA 02	Number Systems and Math Foundations for DSA Interviews
DSA	DSA 03	Number Systems Interview Patterns and Rapid Revision
DSA	DSA 04	Bit Manipulation in Java
DSA	DSA 05	Loop Mastery, Patterns, and Index Calculations
DSA	DSA 06	Arrays and Array Problem-Solving Patterns
DSA	DSA 07	Strings and String Problem-Solving Patterns
DSA	DSA 08	Hashing: Maps, Sets, Frequency, and Prefix State
DSA	DSA 09	Recursion and Backtracking in Java
DSA	DSA 10	Linked Lists: Pointer Reasoning and Mutation
DSA	DSA 11	Stacks, Queues, Deques, and Monotonic Patterns
DSA	DSA 12	Binary Search: Bounds, Answers, and Invariants
DSA	DSA 13	Trees, Binary Search Trees, and Tries
DSA	DSA 14	Heaps, Priority Queues, Selection, and Top-K
DSA	DSA 15	Graphs: Traversal, Ordering, Paths, and Union-Find
DSA	DSA 16	Greedy Algorithms: Recognition, Proofs, and Scheduling
DSA	DSA 17	Dynamic Programming: State, Transitions, and Optimization
System Design	SD 01	Advanced Java F: Design, Backend, Testing, and Security
System Design	SD 02	MySQL for Java Backend Interviews
System Design	SD 03	Hibernate and JPA for Java Backend Interviews
System Design	SD 04	Spring Framework for Java Engineers
System Design	SD 05	Spring Boot for Java Backend Engineers

SEGMENT	BOOK	FOCUSED LEARNING PATH
System Design	SD 06	Spring Data for Java Backend Engineers
System Design	SD 07	MongoDB for Java Backend Interviews
System Design	SD 08	Redis for Java Backend Interviews
System Design	SD 09	Apache Kafka and Spring Kafka
System Design	SD 10	Spring Ecosystem Extensions
System Design	SD 11	Spring AI for Java Engineers
System Design	SD 12	Advanced Java H: Spring Boot and REST APIs
System Design	SD 13	Advanced Java I: Persistence, SQL, JPA, and Caching
System Design	SD 14	Advanced Java J: Distributed Systems and System Design

Roadmap editions identify planned books whose chapter sets will be expanded one at a time. Existing deep-dive books remain available later in each segment, so no published material is displaced.

About the Author

Vinay Reddy Kalluri

Vinay Reddy Kalluri is a senior Java backend engineer, the creator and founding author of the Java SDE-2 Interview Preparation Series, and its Editor-in-Chief and Chief Auditor. His work spans high-throughput microservices, event-driven architecture, resilient APIs, large-scale data processing, cloud delivery, and production performance across healthcare and enterprise systems.

Editorial leadership

As Editor-in-Chief, Vinay owns the learning sequence, scope, voice, and publication decisions. As Chief Auditor, he owns the Java accuracy standard, validation evidence, attribution review, and release-readiness gate. Individual contributors retain visible credit for accepted original work through AUTHORS.md and the public Git history.

What distinguishes his approach is the combination of hands-on system design and operational ownership. He treats timeouts, retries, idempotency, dead-letter recovery, data consistency, SQL behavior, caching, and observability as part of the design rather than afterthoughts. He has also mentored engineers and led modernization, migration, and reliability work across distributed services.

Selected engineering and academic highlights

- More than eight years building Java and Spring Boot services for high-throughput healthcare and enterprise platforms.
- Designed Kafka-based event systems that have processed more than one million events per hour, with explicit idempotency, retry, recovery, and observability controls.
- M.S. in Computer Science from the University of Missouri-Kansas City (3.9/4.0) and M.S. in Software Engineering from VIT University (9.3/10), where he was a Gold Medalist.
- Independent builder of Work Visa Insights, an immigration-data intelligence platform, and Play Together, a published Android application.

Why this series exists

Vinay created this series to turn fragmented interview preparation into a deliberate progression: learn the fundamentals, run the examples, predict behavior, debug failures, explain trade-offs, and only then move into advanced engineering. The books reflect the same principle he applies to production systems: clarity before cleverness, explicit contracts, measurable behavior, and reliable foundations.

Connect: [LinkedIn](#) | [GitHub](#)

Copyright and Publishing Notes

Copyright 2026 Vinay Reddy Kalluri and credited contributors. Open educational edition.

Book prose, exercises, diagrams, and PDFs are licensed under Creative Commons Attribution 4.0 International (CC BY 4.0). Build scripts and source code are licensed under MIT. Individual credit is recorded in AUTHORS.md and Git history.

Repository: <https://github.com/vinayreddykalluri/SDE2-Interview-Handbook>. Attribution does not imply endorsement. Java and related marks are owned by their respective holders.

Examples target Java 21 unless a section states otherwise. Validate release-specific APIs, JVM behavior, security, performance, licensing, and operational requirements before production use.

Series position: Series Index - 3 Segments / 40 Books. The local table of contents and PDF bookmarks navigate within this file; the roadmap and printed filenames navigate across the complete series.

Source provenance: curated from the independent Java SDE-2 master guide plus focused original material. Original master chapter numbers are labeled explicitly when retained in cross-references.